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"From Rookie to Rock"

Learn & Practice
FREE PowerBI pbix

Profitability of a project

Financial UDF pack

Part 1: How it works, financial explanations



Patou Tips #60





Profitability of a project Financial UDF pack (1/3)



About this financial package...

This package of 8 DAX functions is designed to provide access to key performance indicators (KPIs) representative of project profitability, such as IRR, NPV, and payback. It also allows for the calculation of the metrics such as DF, FCF, and DCF.

In order to easily transpose these functions into your projects, and to use these functions as DAX functions, these 8 functions are developed in **UDF (User Defined Function)** mode.

Functions	What it is?
DF.PatouTips	Calculate DF (Discounted Factor) by period
FCF.PatouTips	Calculate FCF (Discounted Free Cash Flow) by period
DCF.PatouTips	Calculate DCF (Discounted Cash Flow) by period
IRR.PatouTips	Calculate IRR (Internal Rate Return) by period
NPV.PatouTips	Calculate NPV (Net Present Value) by period
NPV.Total.PatouTips	Calculate NPV (Net Present Value) at end of period
Payback.Year.PatouTips	Calculate the year of the project payback
Payback.Nb.Year.PatouTips	Calculate the number of year of the project payback

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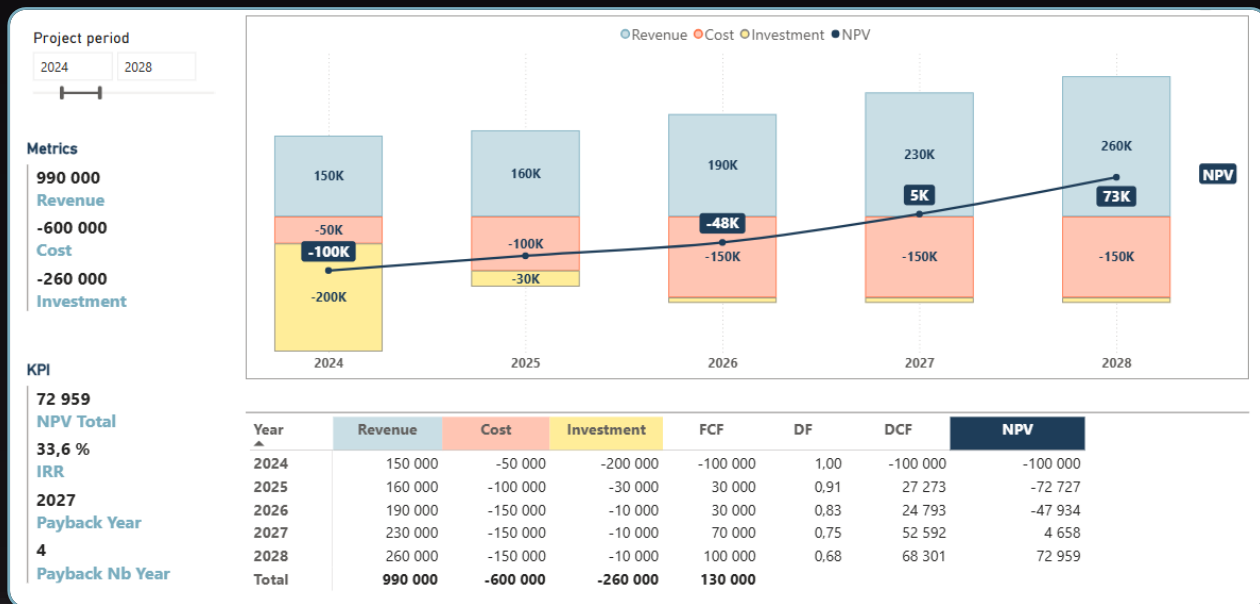
Profitability of a project

Financial UDF pack (2/3)



About this financial package...

💡 Functions are created to be used independently, in whole or in part. Functions are not linked to each other by code. Each function generates an value for each period used to calculate the profitability of a project. These values can be displayed as a matrix or a graph for example.



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Profitability of a project Financial UDF pack (3/3)



About this financial package...

Find all the resources on the Patou Tips GitHub

<https://github.com/PatouTips/Patou-Tips/>

Patou-Tips / #60 Patou Tips (Financial UDF pack - Profitability of a project) /

PatouTips Add files via upload

Name	Last commit message
..	
#60 Patou Tips - Part 1.pdf	Add files via upload
Financial function with UDF (Profitability of a project).pbix	Add files via upload
README.md	Update README.md
Script TMDL - Financial UDF Pack	Profitability of a proje... Add files via upload

Repository

To understand
all about....

PowerBI file
example

To go further...

Script TMDL to transpose functions in your PowerBI
project

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Finance UDF (User Defined Function) Pack

Profitability of a project

Part 1:
How it works?
Financial explanations





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Profitability of a project

How it works? (1/4)

2 ways* to download UDF functions from the Patou Tips GitHub repository, chapter #60:

1 - Open the file "**Script TMDL - Financial UDF Pack - Profitability of a project.txt**", copy the contents of this text file, then paste the entire script into the TMDL view of your Power BI project (see Patou Tips #58 for more information).

 Script TMDL - Financial UDF Pack - Profitability of a project.txt

2 - Go to the "**READ.md**" file and click the copy icon in chapter "3 - Code".

3 - Code

createOrReplace

```
/// Discounted Cash Flow (DCF) is a valuation method that estimates the value  
function 'DCF.Patou.Tips' = ````
```



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Profitability of a project

How it works? (2/4)

 Create a measure that calls one of the "Patou.Tips" functions, for example, write the word "Patou".

The function-specific instructions are also summarized in the pop-up window that appears when the measure is inserted.

```
NPV = NPV.Patou.Tips([Revenue]
```

```
NPV.Patou.Tips(Revenue_measure: expr, Cost_measure: expr,  
Investment_measure: expr, Year_date_of_calendar_table: anyref,  
Discounted_rate: decimal)
```

Net Present Value (NPV). This is the cumulative Discounted Cash Flow (DCF), for each period of the project. NPV evaluates the profitability of an investment by comparing the present value of expected future cash flows to the initial investment. Put Costs and Investment in negative value. The Discounted Rate is a decimal value representing a percentage. For example, 0.101 for 10.1%.



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Profitability of a project

How it works? (3/4)

Download the Power BI file which will give you a complete example of its use in a financial project.

 Financial function with UDF (Profitability of a project).pbix

The dataset uses in this Patou Tips comes from the fictitious Parisian company "IceCream' Macaron," taken from my book "Business Intelligence".

Year	Revenue	Cost	Investment
2024	150 000 €	-50 000 €	-200 000 €
2025	160 000 €	-100 000 €	-30 000 €
2026	190 000 €	-150 000 €	-10 000 €
2027	230 000 €	-150 000 €	-10 000 €
2028	260 000 €	-150 000 €	-10 000 €
2029	275 000 €	-150 000 €	-10 000 €
2030	300 000 €	-150 000 €	-10 000 €
2031	330 000 €	-150 000 €	-10 000 €
2032	360 000 €	-150 000 €	-10 000 €



→ This project represents the launch of a new store for the company "IceCream' Macaron." It involves a significant initial investment and residual investments for possible improvements to the store structure.

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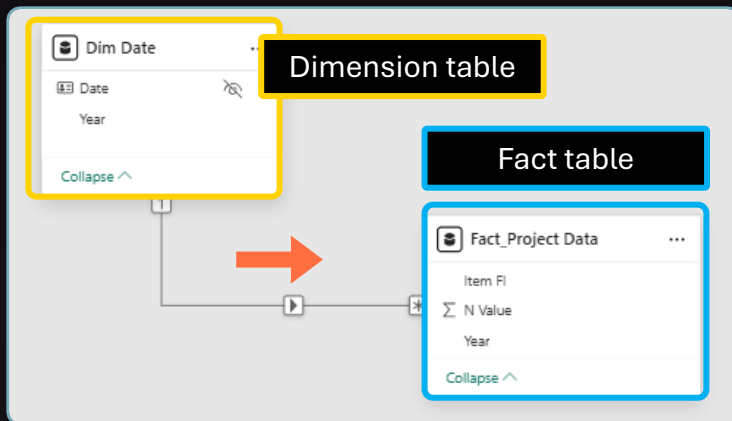
Profitability of a project

How it works? (4/4)

To use the UDF of this financial pack, you need at least 3 measures :

- ✓ Revenue,
- ✓ Cost,
- ✓ Investment
- ✓ And a calendar table with the "Year" field at least

Note: Cost and Investment measures be in negative values.



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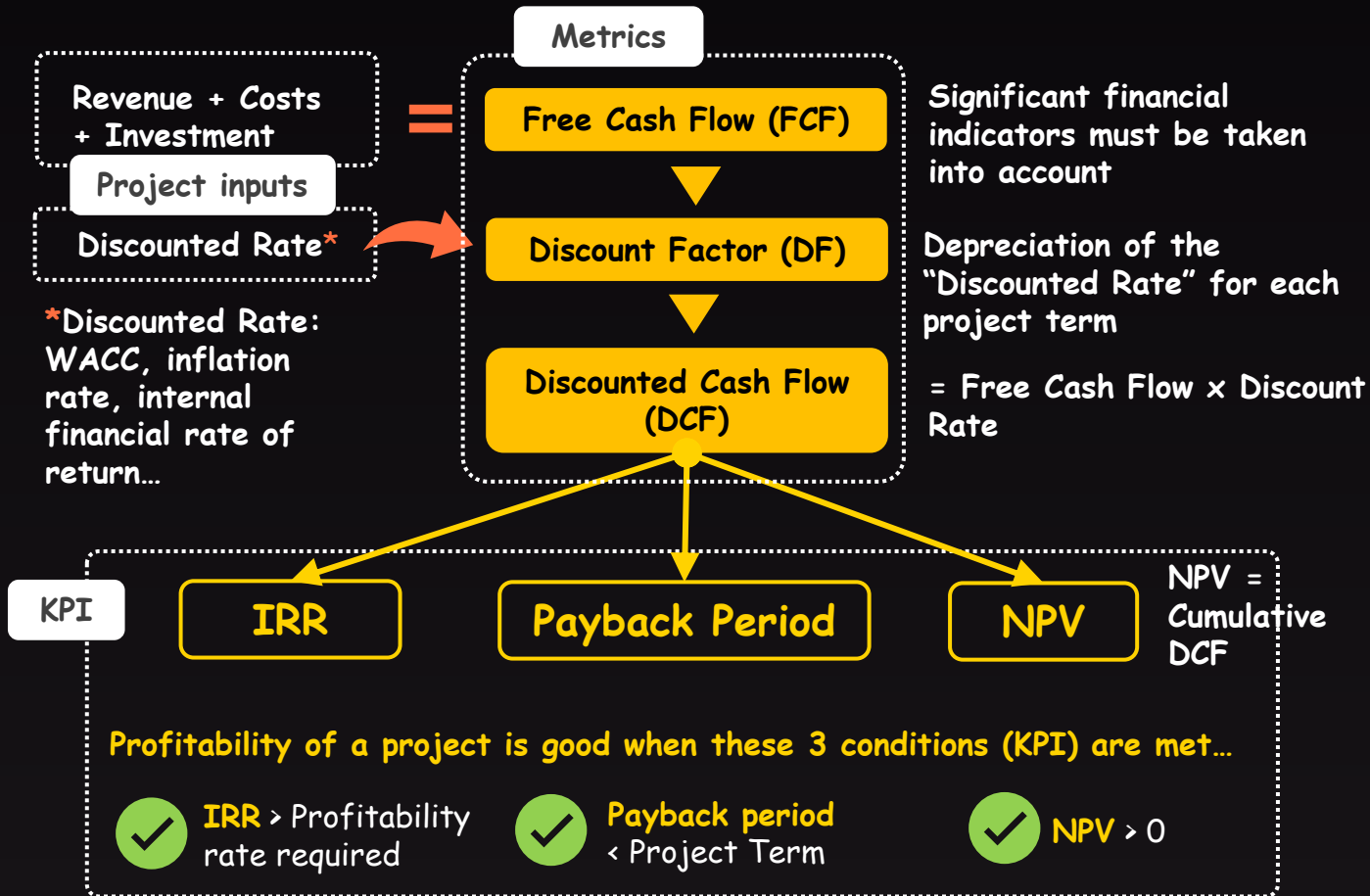
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Profitability of a project

Financial explanations (1/3)



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1

Profitability of a project

Financial explanations (2/3)

The **profitability of a project** is essential for making informed investment decisions. This assessment, which incorporates the time value of money, the ****discount factor (DF)****, involves using tools such as:

The **Net Present Value (NPV)** calculates the difference between the present value of future cash flows and the initial investment. A positive NPV indicates that the project generates more value than its cost, thus signaling a profitable opportunity. Conversely, a negative NPV suggests an unviable project.

The **Internal Rate of Return (IRR)** represents the **discount factor (DF)** that makes the **NPV equal to zero**. It allows you to compare the project's profitability to a minimum required rate of return (cost of capital). An IRR higher than this rate means the project is attractive.



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Profitability of a project

Financial explanations (3/3)

An indicator such as the **payback period** complements this analysis by assessing the time required to obtain a return on investment.

In summary, these tools offer a clear view of risks and opportunities, helping investors and managers to prioritize the most profitable projects, in line with their strategic objectives.

Coming soon, in 2026 / 2027!



Patrice Fayard

Business Intelligence WORKBOOK

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- ✓ 5 Cheat Sheet
- ✓ 100 Power Patou Tips
- ✓ Hacking & Workshops



Over 500 pages to
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tutorials and resources

Project version shown

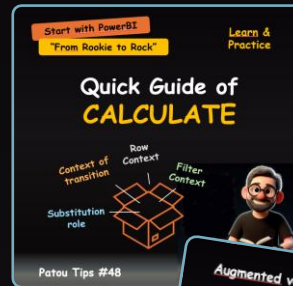


Learn and practice

Find past issues of "Patou Tips" and download resources to practice on GitHub



Patou Tips #22
PowerBI projects with Teams



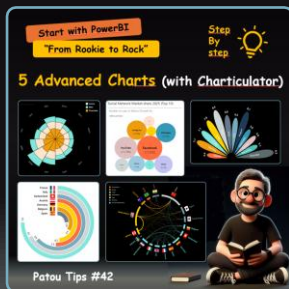
Patou Tips #48
Quick Guide of CALCULATE



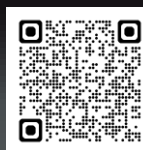
Patou Tips #23
Quick Guide Visualization & Storytelling



Patou Tips #25
Start with PowerBI: The complete Guide



Patou Tips #42
5 Advanced Charts (with Charticulator)



Resources on GitHub
<https://github.com/PatouTips/Patou-Tips>

Patou Tips



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